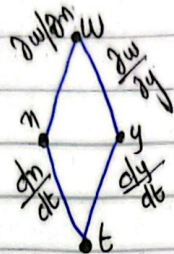


Exercise # 14.4

Chain Rule: One Independent variable

Q1- $w = n^2 + y^2$, $n = \cos t$, $y = \sin t$; $t = \pi$

Sol:



$$\frac{dw}{dt} = \frac{\partial w}{\partial n} \cdot \frac{dn}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt}$$

= Partial derivative with Respect to n & y

$$\frac{\partial w}{\partial n} = 2n \quad \frac{\partial w}{\partial y} = 2y$$

Derivative of n & y

$$\frac{dn}{dt} = -\sin t \quad \frac{dy}{dt} = \cos t$$

put into formula

$$\frac{dw}{dt} = 2n \cdot (-\sin t) + 2y \cdot (\cos t)$$

$$\frac{dw}{dt} = -2n \sin t + 2y \cos t$$

put the value of n & y

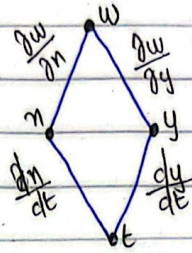
$$\frac{dw}{dt} = -2(\cos t) \sin t + 2(\sin t) (\cos t)$$

$$\frac{dw}{dt} = -2 \cos t \sin t + 2 \sin t \cos t$$

$$= 0$$

Q2- $w = n^2 + y^2$, $n = \cos t + \sin t$, $y = \cos t - \sin t$; $t=0$

Sol:



$$\frac{dw}{dt} = \frac{\partial w}{\partial n} \cdot \frac{dn}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt}$$

Partial derivative with respect to n and y .

$$\frac{\partial w}{\partial n} = 2n$$

$$\frac{\partial w}{\partial y} = 2y$$

Derivative of n & y

$$\frac{dn}{dt} = -\sin t + \cos t$$

$$\frac{dy}{dt} = -\sin t - \cos t$$

put into formula

$$\frac{dw}{dt} = 2n \cdot (-\sin t + \cos t) + 2y \cdot (-\sin t - \cos t)$$

Interchange

$$\frac{dw}{dt} = 2n (\cos t - \sin t) - 2y (\sin t + \cos t)$$

$$\frac{dw}{dt} = 2(\cos t + \sin t)(\cos t - \sin t) - 2(\cos t - \sin t)(\sin t + \cos t)$$

$$\frac{dw}{dt} = 2(\cos^2 t - \cancel{\cos t \sin t} + \cancel{\sin t \cos t} - \sin^2 t) - 2(\cancel{\cos t \sin t} + \cos^2 t - \sin^2 t - \cancel{\cos t \sin t})$$

$$\frac{dw}{dt} = 2(\cos^2 t - \sin^2 t) - 2(\cos^2 t - \sin^2 t)$$

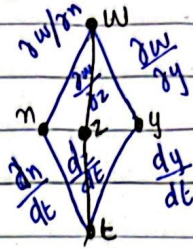
$$= 0$$

Q3- $w = \frac{n}{z} + \frac{y}{z}$, $n = \cos^2 t$, $y = \sin^2 t$, $z = 1/t$; $t = 3$

Sol:-

$$w = \frac{n+y}{z}$$

Now Sol:-



$$\frac{dw}{dt} = \frac{\partial w}{\partial n} \cdot \frac{dn}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$$

Partial derivative of n, y, z

$$\frac{\partial w}{\partial n} = \frac{1}{z}, \quad \frac{\partial w}{\partial y} = \frac{1}{z}, \quad \frac{\partial w}{\partial z} = -\frac{n+y}{z^2}$$

Derivative of n, y, z

$$\frac{dn}{dt} = -2 \cos t \sin t, \quad \frac{dy}{dt} = 2 \sin t \cos t, \quad \frac{dz}{dt} = -\frac{1}{t^2}$$

put all of these values into formula

$$\frac{dw}{dt} = \frac{1}{z} \cdot (-2 \cos t \sin t) + \frac{1}{z} \cdot (2 \sin t \cos t) + \frac{-(n+y)}{z^2} \cdot -\frac{1}{t^2}$$

$$\frac{dw}{dt} = \cancel{-\frac{2 \cos t \sin t}{z}} + \cancel{\frac{2 \sin t \cos t}{z}} + \frac{n+y}{z^2 \cdot t^2}$$

$$\frac{dw}{dt} = \frac{n+y}{z^2 t^2}$$

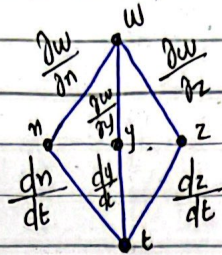
put values

$$\frac{dw}{dt} = \frac{\cos^2 t + \sin^2 t}{\left(\frac{1}{t}\right)^2 \cdot t^2} \Rightarrow \frac{\cos^2 t + \sin^2 t}{\frac{1}{t} \cdot t^2}$$

$$\frac{dw}{dt} = \cos^2 t + \sin^2 t$$

04- $w = \ln(n^2 + y^2 + z^2)$, $n = \cos t$, $y = \sin t$, $z = 4\sqrt{t}$; $t = 3$

Sol:-



$$\frac{dw}{dt} = \frac{\partial w}{\partial n} \cdot \frac{dn}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$$

Partial derivative of n, y, z

$$\frac{\partial w}{\partial n} = \frac{1}{n^2 + y^2 + z^2} \cdot \frac{\partial}{\partial n}, \quad \frac{\partial w}{\partial y} = \frac{1}{n^2 + y^2 + z^2} \cdot \frac{\partial}{\partial y}, \quad \frac{\partial w}{\partial z} = \frac{1}{n^2 + y^2 + z^2} \cdot \frac{\partial}{\partial z}$$

Derivative of n, y, z

$$\frac{dn}{dt} = -\sin t, \quad \frac{dy}{dt} = \cos t, \quad \frac{dz}{dt} = \frac{4}{2\sqrt{t}}$$

Put these into formula

$$\frac{dw}{dt} = \frac{\partial w}{\partial n} \cdot (-\sin t) + \frac{\partial w}{\partial y} (\cos t) + \frac{\partial w}{\partial z} \cdot \frac{4}{2\sqrt{t}}$$

$$\frac{dw}{dt} = \frac{-\frac{\partial w}{\partial n} \sin t}{n^2 + y^2 + z^2} + \frac{\frac{\partial w}{\partial y} \cos t}{n^2 + y^2 + z^2} + \frac{8 \frac{\partial w}{\partial z} \sqrt{t}}{(n^2 + y^2 + z^2)\sqrt{t}}$$

Put these values

$$\frac{dw}{dt} = \frac{-\frac{\partial w}{\partial n} (\cos t) \sin t}{(\cos^2 t) + (\sin^2 t) + (4\sqrt{t})^2} + \frac{\frac{\partial w}{\partial y} (\sin t) \cos t}{(\cos^2 t) + (\sin^2 t) + (4\sqrt{t})^2} + \frac{8(4\sqrt{t}) 4(4\sqrt{t})}{(\cos^2 t + \sin^2 t + (4\sqrt{t})^2)\sqrt{t}}$$

$$\frac{dw}{dt} = \frac{-\frac{\partial w}{\partial n} \cos(3) \sin(3)}{\cos^2(3) + \sin^2(3) + 16(3)} + \frac{\frac{\partial w}{\partial y} \sin(3) \cos(3)}{\cos^2(3) + \sin^2(3) + 16(3)} + \frac{8(4\sqrt{3}) 4(4\sqrt{3})}{(\cos^2(3) + \sin^2(3) + 16(3))\sqrt{3}}$$

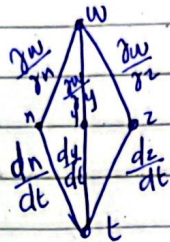
$$\frac{dw}{dt} = \frac{\cancel{8(4\sqrt{t})} 4(4\sqrt{t})}{(\cos^2 t + \sin^2 t + 16(\sqrt{t})^2) \sqrt{t}}$$

$$\frac{dw}{dt} = \frac{\cancel{8(4\sqrt{t})} 4(4\sqrt{t})}{(1+16t)\sqrt{t}} \Rightarrow \frac{16}{1+16t}$$

$$\frac{dw}{dt} = \frac{16}{1+16(3)} = \frac{16}{1+48} \Rightarrow \frac{16}{49}$$

Q5- $w = ye^n - \ln z$, $n = \ln(t^2 + 1)$, $y = \tan^{-1} t$, $z = e^t$
 $t = 1$

Solution:



$$\frac{dw}{dt} = \frac{\partial w}{\partial n} \cdot \frac{dn}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$$

Partial derivative

$$\frac{\partial w}{\partial n} = ye^n , \quad \frac{\partial w}{\partial y} = e^n , \quad \frac{\partial w}{\partial z} = -\frac{1}{z}$$

Derivative

$$\frac{dn}{dt} = \frac{1}{t^2 + 1} \cdot 2t , \quad \frac{dy}{dt} = \frac{1}{1 + t^2} , \quad \frac{dz}{dt} = e^t$$

Put into formula

$$\frac{dw}{dt} = ye^n \cdot \frac{2t}{t^2 + 1} + e^n \cdot \frac{1}{1 + t^2} + \left(-\frac{1}{z}\right) \cdot e^t$$

$$\frac{dw}{dt} = \frac{4ye^n t}{t^2 + 1} + \frac{e^n}{1 + t^2} - \frac{e^t}{z}$$

put n, y, z

$$= \frac{4(\tan^{-1} t) e^{\ln(t^2 + 1)}}{1^2 + 1} \cdot 1 + \frac{e^{\ln(t^2 + 1)}}{1 + 1^2} - \frac{e^1}{e^1}$$

$$= \frac{4(\tan^{-1} t) e^{\ln(t^2 + 1)}}{2} + \frac{e^{\ln(t^2 + 1)}}{2} - 1$$

$$= 2(\tan^{-1} t) e^{\ln(t^2 + 1)} + \frac{1}{2} e^{\ln(t^2 + 1)} - 1$$

put the m, y, z values.

$$\frac{dw}{dt} = \frac{4ye^y}{t^2+1} + \frac{2e^y}{t^2+1} - \frac{1}{z} e^t$$

$$= \frac{4t(\tan^{-1}t)e^{\ln(t^2+1)}}{t^2+1} + \frac{2(e^{\ln(t^2+1)})}{t^2+1} - e^{-t} \cdot e^t$$

$$e^{\ln(t^2+1)} = t^2+1$$

$$= \frac{4t(\tan^{-1}t)(\cancel{t^2+1})}{\cancel{t^2+1}} + \frac{2(\cancel{t^2+1})}{\cancel{t^2+1}} - \frac{e^{-t}}{e^t}$$

$$= 4t(\tan^{-1}t) + 2 - 1$$

$$= 4t(\tan^{-1}t) + 1$$

Put the value of t

$$\frac{dw}{dt} = 4(1)(\tan^{-1}(1)) + 1$$

$$= 4(\tan^{-1}(1)) + 1$$

$$\frac{dw}{dt} = \cancel{4} \frac{\pi}{\cancel{4}} + 1$$

$$\frac{dw}{dt} = \pi + 1$$

Q6-

$$w = z - \sin ny, \quad n = t, \quad y = \ln t, \quad z = e^{t-1} = t-1$$

Sol:-

$$\frac{dw}{dt} = \frac{\partial w}{\partial n} \cdot \frac{dn}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dt}$$

Partial derivative n, y, z

$$\frac{\partial w}{\partial n} = -y \cos ny, \quad \frac{\partial w}{\partial y} = -n \cos ny, \quad \frac{\partial w}{\partial z} = 1$$

Derivative of n, y, z

$$\frac{dn}{dt} = 1, \quad \frac{dy}{dt} = \frac{1}{t}, \quad \frac{dz}{dt} = e^{t-1}$$

Put into formula

$$\frac{dw}{dt} = -y \cos ny (1) + -n \cos ny \cdot \frac{1}{t} + 1 \cdot e^{t-1}$$

$$\frac{dw}{dt} = -y \cos ny - \frac{n \cos ny}{t} + e^{t-1}$$

Put n y z value

$$= -(\ln t) \cos(t)(\ln t) - \frac{t \cos(t \ln t)}{t} + e^{t-1}$$

$$= -\ln t \cos(t \ln t) - \cos(t \ln t) + e^{t-1}$$

Common

$$= -\cos(t \ln t) (\ln t + 1) + e^{t-1}$$

put $t = 1$

$$= -\cos(\ln(1)) (\ln(1) + 1) + e^{1-1}$$

$$= -\cos(0) (0 + 1) + 1$$

$$= -\cos(0) (1) + 1$$

$$= -1 + 1$$

$$= 0$$

Chain rule: Two and Three Independent variables

$$D7- z = 4e^n \ln y, \quad n = \ln(u \cos v) \quad y = u \sin v$$
$$(u, v) = (z, \pi/4)$$

Solution:

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial n} \cdot \frac{\partial n}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u} \rightarrow \text{All Partial value bcz 2 independent variables.}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial n} \cdot \frac{\partial n}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

Partial derivative of $\partial z / \partial u$.

$$\frac{\partial z}{\partial n} = 4e^n \ln y, \quad \frac{\partial n}{\partial u} = \frac{1}{u \cos v} \cdot \cos v \Rightarrow \frac{1}{u}, \quad \frac{\partial z}{\partial y} = \frac{4e^n}{y}, \quad \frac{\partial y}{\partial u} = \sin v$$

Put into formula.

$$\frac{\partial z}{\partial u} = 4e^n \ln y \cdot \frac{1}{u} + \frac{4e^n}{y} \cdot \sin v \Rightarrow \frac{4e^n \ln y}{u} + \frac{4e^n \sin v}{y}$$

$$\frac{\partial z}{\partial u} = \frac{4e^{\ln(u \cos v)}}{u} \ln(u \sin v) + \frac{4e^{\ln(u \cos v)}}{u \sin v} \sin v$$

$$= \frac{4(\cancel{u} \cos v)}{\cancel{u}} \ln(u \sin v) + 4 \frac{(\cancel{u} \cos v)}{\cancel{u} \cancel{\sin v}} \sin v$$

$$= 4 \cos v \ln(u \sin v) + 4 \cos v$$

Now put n and y values into function

$$z = 4e^n \ln y$$
$$= 4e^{\ln(u \cos v)} \ln(u \sin v)$$

$$= 4(u \cos v) \times \ln(u \sin v)$$

$$\frac{\partial z}{\partial u} = 4 \cos v \times \frac{\partial}{\partial u} (u \cdot \ln(u \sin v))$$

Apply Product Rule.

$$= 4 \cos v \left[\cancel{u} \cdot \frac{1}{\cancel{u} \sin v} \times \sin v + (1) \cdot \ln(u \cdot \sin v) \right]$$

$$\frac{\partial z}{\partial u} = 4 \cos v + 4 \cos v (\ln u \cdot \sin v)$$

Now Solve with Respect to v

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

Partial derivative

$$\frac{\partial z}{\partial u} = 4e^u \ln y, \frac{\partial z}{\partial v} = \frac{1}{u \cos v} \cdot u(-\sin v), \frac{\partial z}{\partial y} = \frac{4e^u}{y}, \frac{\partial y}{\partial v} = u \cos v$$

Put it into formula

$$\frac{\partial z}{\partial v} = 4e^u \ln y \cdot -\frac{\sin v}{\cos v} + \frac{4e^u}{y} \cdot u \cos v$$

$$\frac{\partial z}{\partial v} = \cancel{4e^{u \ln(u \cos v)}} \cdot -\frac{\sin v}{\cos v} + \cancel{4e^{u \ln(u \cos v)}} \cdot u \cos v$$

$$= \cancel{-4u \sin v} + \cancel{\frac{4(u \cos v)}{u \sin v}} \cdot u \cos v$$

$$\frac{\partial z}{\partial v} = \cancel{4e^{u \ln(u \cos v)}} \cdot \ln(u \sin v) \cdot -\frac{\sin v}{\cos v} + \cancel{4e^{u \ln(u \cos v)}} \cdot u \cos v$$

$$= -4u \sin v \cdot \ln(u \sin v) + \frac{4u \cos^2 v}{u \sin v} \Rightarrow -4u \sin v \cdot \ln(u \sin v) + 4u \cos^2 v \sin v$$

Now put m & y z into original function

$$z = 4e^m \ln y$$

$$\begin{aligned} z &= 4e^{\ln(u \cos v)} \cdot \ln(u \sin v) \\ &= 4(u \cos v) \cdot \ln(u \sin v) \\ &\quad \cancel{4e^{\ln(u \sin v)}} \cdot \ln(u \sin v) \end{aligned}$$

$$\frac{\partial z}{\partial v} = 4u (\cos v) \cdot \ln(u \sin v)$$

Apply Product Rule

$$\frac{\partial z}{\partial u} = 4u \left[\cos v \times \frac{1}{u \sin v} \cdot \cancel{u \cos v} + (-\sin v) \cdot \ln(u \sin v) \right]$$

$$\frac{4u \cos^2 v}{\sin v} - 4u \sin v \cdot \ln(u \sin v)$$

Both Answer with respect to v is Same.

$$09-w = ny + yz + mz, \quad n = u+v, \quad y = u-v, \quad z = uv \quad (u,v) = (1/2, 2)$$

Solution:

$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial n} \cdot \frac{dn}{du} + \frac{\partial w}{\partial y} \cdot \frac{dy}{du} + \frac{\partial w}{\partial z} \cdot \frac{dz}{du}$$

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial n} \cdot \frac{dn}{dv} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dv} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dv}$$

Partial derivative.

$$\frac{\partial w}{\partial n} = y + z, \quad \frac{\partial w}{\partial y} = n + z, \quad \frac{\partial w}{\partial z} = y + n$$

Derivative

$$\frac{dn}{du} = 1, \quad \frac{dy}{du} = 1, \quad \frac{dz}{du} = v$$

Put into formula

$$\frac{\partial w}{\partial z} = y + z \times (1) + (n + z)1 + (y + n)v$$

$$\frac{\partial w}{\partial z} = y + z + n + z + yv + nv \Rightarrow y + 2z + n + yv + nv$$

Put n, y, z values.

$$\frac{\partial w}{\partial z} = (u-v) + 2(uv) + (u+v) + (u-v)v + (u+v)v$$

$$= u \cancel{-v} + 2uv + u \cancel{+v} + uv \cancel{-v^2} + uv \cancel{+v^2}$$

$$= 2u + 2uv + 2uv$$

$$\frac{\partial w}{\partial z} = 2u + 4uv$$

at

Now Solve with Respect to $\partial w / \partial v$

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial n} \cdot \frac{dn}{dv} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dv} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dv}$$

Partial Derivative

$$\frac{\partial w}{\partial n} = y + z, \quad \frac{\partial w}{\partial y} = n + z, \quad \frac{\partial w}{\partial z} = y + n$$

Derivative of n, y, z

$$\frac{dn}{dv} = 1, \quad \frac{dy}{dv} = -1, \quad \frac{dz}{dv} = u$$

Put it into formula

$$\frac{\partial w}{\partial v} = (y+z) \cdot (1) + (n+z) \cdot (-1) + (y+n)u$$

$$= y + z - n - z + yu + zu$$

$$= y - n + yu + zu$$

$$= (u-v) - (u+v) + (u-v)u + (u+v)u$$

$$= u - v - u - v + u^2 - uv + u^2v + u^2 + uv$$

$$= -2uv + u^2 + u^2v + u^2 + uv$$

$$= -2uv + u^2 + u^2v$$

$$= -2v + u^2 + u^2$$

$$= -2v + 2u^2$$

$$10- w = \ln(x^2 + y^2 + z^2), \quad x = ue^v \sin u, \quad y = ue^v \cos u, \quad z = ue^v$$

$$(u, v) = (-2, 0)$$

Solution:

$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x} \cdot \frac{dx}{du} + \frac{\partial w}{\partial y} \cdot \frac{dy}{du} + \frac{\partial w}{\partial z} \cdot \frac{dz}{du}$$

$$\frac{\partial w}{\partial v} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dv} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dv} + \frac{\partial w}{\partial z} \cdot \frac{dz}{dv}$$

Partial Derivative

$$\frac{\partial w}{\partial x} = \frac{1}{x^2 + y^2 + z^2} \cdot x, \quad \frac{\partial w}{\partial y} = \frac{1}{x^2 + y^2 + z^2} \cdot y, \quad \frac{\partial w}{\partial z} = \frac{1}{x^2 + y^2 + z^2} \cdot z$$

Derivative

$$\frac{dx}{du} = e^v \cos u, \quad \frac{dy}{du} = -e^v \sin u, \quad \frac{dz}{du} = e^v$$

Put into formula

$$\frac{\partial w}{\partial u} = \frac{x \cdot e^v \cos u}{x^2 + y^2 + z^2} + \frac{y \cdot (-e^v \sin u)}{x^2 + y^2 + z^2} + \frac{z \cdot e^v}{x^2 + y^2 + z^2}$$

$$\frac{\partial w}{\partial u} = \frac{x e^v \cos u}{x^2 + y^2 + z^2} - \frac{y e^v \sin u}{x^2 + y^2 + z^2} + \frac{z e^v}{x^2 + y^2 + z^2}$$

$$11) u = \frac{p-q}{q-r}, \quad p = n+y+z, \quad q = n-y+z, \quad r = n+y-z; \quad (n, y, z) = (\sqrt{3}, 2, 1)$$

Apply Quotient Rule

Solution:

Partial Derivative

$$\frac{\partial u}{\partial p} = \frac{(q-r)(1) - (p-q)(0)}{(q-r)^2} \Rightarrow \frac{q-r}{(q-r)^2} \Rightarrow \frac{1}{q-r}$$

$$\frac{\partial u}{\partial q} = \frac{(q-r)(-1) - (p-q)1}{(q-r)^2} \Rightarrow \frac{-q+r-p+q}{(q-r)^2} \Rightarrow \frac{r-p}{(q-r)^2}$$

$$\frac{\partial u}{\partial r} = \frac{(q-r)(0) - (p-q)(-1)}{(q-r)^2} \Rightarrow \frac{-(-p+q)}{(q-r)^2} \Rightarrow \frac{p-q}{(q-r)^2}$$

Derivative of n, y, z

$$\frac{dp}{dn} = 1, \quad \frac{dq}{dn} = 1, \quad \frac{dr}{dn} = 1, \quad \frac{dp}{dy} = 1, \quad \frac{dq}{dy} = -1, \quad \frac{dr}{dy} = 1$$

$$\frac{dp}{dz} = 1, \quad \frac{dq}{dz} = 1, \quad \frac{dr}{dz} = -1$$

put it into formula with respect to n

$$\frac{\partial u}{\partial n} = \frac{\partial u}{\partial p} \cdot \frac{dp}{dn} + \frac{\partial u}{\partial q} \cdot \frac{dq}{dn} + \frac{\partial u}{\partial r} \cdot \frac{dr}{dn}$$

$$= \frac{1}{q-r} \cdot 1 + \frac{r-p}{(q-r)^2} \cdot 1 + \frac{p-q}{(q-r)^2} \cdot 1$$

$$= \frac{1}{q-r} + \frac{r-p}{(q-r)^2} + \frac{p-q}{(q-r)^2} \Rightarrow \frac{(q-r) + (r-p) + (p-q)}{(q-r)^2}$$

$$= \frac{q-r+r-p+p-q}{(q-r)^2} \Rightarrow \frac{1}{(q-r)^2}$$

put it into formula with Respect to y

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial p} \frac{dp}{dy} + \frac{\partial u}{\partial q} \frac{dq}{dy} + \frac{\partial u}{\partial r} \frac{dr}{dy}$$

$$= \frac{1}{q-r} \cdot 1 + \frac{r-p}{(q-r)^2} \cdot (-1) + \frac{p-q}{(q-r)^2} \cdot 1$$

$$= \frac{1}{q-r} - \frac{r-p}{(q-r)^2} + \frac{p-q}{(q-r)^2} \Rightarrow \frac{(q-r) - (r-p) + (p-q)}{(q-r)^2}$$

$$\frac{q-r-r+p+p-q}{(q-r)^2} \Rightarrow \frac{-2r+2p}{(q-r)^2} \Rightarrow \frac{2p-2r}{(q-r)^2}$$

put it into formula with Respect to z

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial p} \frac{dp}{dz} + \frac{\partial u}{\partial q} \frac{dq}{dz} + \frac{\partial u}{\partial r} \frac{dr}{dz}$$

$$= \frac{1}{q-r} \cdot 1 + \frac{r-p}{(q-r)^2} \cdot 1 + \frac{p-q}{(q-r)^2} \cdot (-1)$$

$$\frac{1}{q-r} + \frac{r-p}{(q-r)^2} - \frac{p-q}{(q-r)^2} \Rightarrow \frac{(q-r) + (r-p) - (p-q)}{(q-r)^2}$$

$$\frac{q-r+r-p-p+q}{(q-r)^2} \Rightarrow \frac{2q-2p}{(q-r)^2}$$

$$12) u = e^{qr} \sin^{-1} p, \quad p = \sin n, \quad q = z^2 \ln y, \quad r = 1/z, \quad (n, y, z) \\ \cdot (\pi/4, 1/2, -1/2)$$

Solution:

Partial Derivative

$$\frac{\partial u}{\partial p} = \frac{e^{qr}}{\sqrt{1-p^2}}, \quad \frac{\partial u}{\partial q} = r e^{qr} \sin^{-1} p, \quad \frac{\partial u}{\partial r} = q e^{qr} \sin^{-1} p$$

Partial Derivative of p, q, r

$$\frac{\partial p}{\partial n} = \cos n, \quad \frac{\partial p}{\partial y} = 0, \quad \frac{\partial p}{\partial z} = 0, \quad \frac{\partial q}{\partial y} = \frac{z^2}{y}, \quad \frac{\partial q}{\partial z} = -\frac{1}{z^2}, \quad \frac{\partial r}{\partial y} = 0$$

$$\frac{\partial p}{\partial z} = 0, \quad \frac{\partial q}{\partial z} = \frac{\partial z^2 \ln y}{\partial z} = 2z \ln y, \quad \frac{\partial r}{\partial z} = -\frac{1}{z^2}$$

put these into formula.

$$\frac{\partial u}{\partial n} = \frac{\partial u}{\partial p} \cdot \frac{\partial p}{\partial n} + \frac{\partial u}{\partial q} \cdot \frac{\partial q}{\partial n} + \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial n}$$

$$\frac{\partial u}{\partial n} = \frac{e^{qr}}{\sqrt{1-p^2}} \cdot \cos n + r e^{qr} \sin^{-1} p \cdot 0 + q e^{qr} \sin^{-1} p \cdot 0$$

$$\frac{\partial u}{\partial n} = \frac{e^{qr}}{\sqrt{1-p^2}} \cdot \cos n$$

Put these into formula with respect to y

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial p} \cdot \frac{\partial p}{\partial y} + \frac{\partial u}{\partial q} \cdot \frac{\partial q}{\partial y} + \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y}$$

$$= \frac{e^{qr}}{\sqrt{1-p^2}} \cdot 0 + r e^{qr} \sin^{-1} p \cdot \frac{z^2}{y} + q e^{qr} \sin^{-1} p \cdot 0$$

$$\frac{\partial u}{\partial y} = r e^{qr} \sin^{-1} p \cdot \frac{z^2}{y}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial p} \cdot \frac{\partial p}{\partial z} + \frac{\partial u}{\partial q} \cdot \frac{\partial q}{\partial z} + \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial z}$$

$$= \frac{e^{qr}}{\sqrt{1-p^2}} \cdot 0 + re^{qr} \sin^{-1} p \cdot 2z \ln y + qe^{qr} \sin^{-1} p \cdot -\frac{1}{z^2}$$

$$= re^{qr} \sin^{-1} p \times 2z \ln y - \frac{qe^{qr} \sin^{-1} p}{z^2}$$

$$e^{qr} \sin^{-1} p \left(2rz \ln y - \frac{q}{z^2} \right)$$